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Analysing Online Vacancy and Skills Demand using Text Mining

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Abstract. The issue of graduates' employability has become a major concern in many countries as the number of graduates keep increasing over the years. Fresh graduates are experiencing difficulties in finding meaningful employment as the growth in hiring has slowed in the recent period. With the impact of fourth industrial revolution, today graduates face a world transformed by technology. The role of universities in higher education become more challenging in shaping more digital technology graduates with the right skills and knowledge. Problems can be seen at both management level of higher education in producing graduates who can meet the needs of industry while industry is also facing the problem of finding skilled graduates who suit their needs. To examine the work skills demanded from Industry of Malaysian, this study proposed a text mining approach from 465 unstructured data of analyst job vacancies in classifying job titles and extracting vacancy and skills from job descriptions. The methodology involved are text retrieval, text pre-processing, word cloud corpus generation and text clustering. The findings indicate that the selected job posts emphasize more on vacancies information rather than skills needed which further shows the gap existence between the skills demand and supply.

1. Introduction

The problem of graduates' employability remains a national continuing issue and putting pressure for universities to improve graduate employment outcomes. According to Malaysian Ministry of Education's Graduate Tracer Study, of the 273,373 graduates in 2015, a large number hold Bachelor's Degree and Diploma, 45% and 43% of all graduates, respectively. Among all graduates, 53% were reported to have started working, 18% chose to pursue further studies, and 24% of graduates were still unemployed. The high level of youth unemployment is caused by too many people graduate from university with a qualification that does not get them a job. Thus, higher education plays an important role in providing a trained workforce. Among several key policy questions being raised are the types of jobs being created and the readiness of the human capital base, and measures to enhance matching in the labour market and alleviate information asymmetry on industry skill needs. However, graduates are not aware of specific competences of job descriptions from the industries that evidence about the skills needed. Minimalizing mismatch between skills needs and supply is now become a concern of policy makers around the world.

Most of the studies investigate the attributes that contributes to successful employment of graduates [1-3] and identifies important competencies for improving graduate employability [4]. Though more parameters have been mentioned, there is still an issue of mismatch between the real data of qualification and skills needed by the industry that is not addressed thoroughly. Currently, the online labour market

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is a powerful force and offers great insight into selected aspects of employment. A practical approach to analysing skills demand from online vacancies is made possible through text mining which able to retrieve and analyse the content of job advertisements. In this paper, vacancy and skills analysis will be performed on online job posts in order to determine the vacancy and skill sets needed by the industry particularly in this case, analyst job vacancies. Therefore, the objectives of this study are 1) to scrap specific competences from job descriptions and 2) to segment the job descriptions through text clustering. A brief review of the literature is presented on approaches and techniques of data mining and text mining. Next, the steps taken for text mining approach are discussed. Then follow with the results of the analysis and finally discussed the implications for the future of research on reducing the gap between skill demands and supply.

2. Related works

The persistence of the graduate unemployment problem warrants the need for predicting factors that are associated with low employability graduates from higher institutions. This issue can be solved by developing a predictive model to predict potential graduates' employability. There were a wide range of prediction models that has been developed by previous researcher including Probit and Tobit models [5] during early twenties. The models developed only used historical data on graduates' employability traditional parameter such as socio-economic conditions, academic performance, ethnicity and type of degree. However, recently as the world transformed by technology with availability of large amount of data, the development of the prediction models in predicting graduates' employability is moving from the traditional statistical model to data mining approach via machine learning [1, 6-8]. More parameters were used to predict graduates' employability that includes additional skill such as cognitive, higher order thinking, technical, personal, practical, generic and others.

Jantawan & Tsai [8] develop a model to predict employees' performance by applying data mining techniques specifically Bayesian method and the Tree method. Work done by [9] use actual data from the graduates themselves after six months of graduation in order to identify attributes that influenced graduates' employability in profiling them as employed, unemployed or others using various algorithms under Bayes and decision methods. A study by Daud et al. [10] predict whether a student will be able to complete his degree or not through learning analytics, discriminative and generative classification models. Attributes such as family expenditures and student's personal information also influenced on student performance prediction. On the other hand, Thakar & Mehta [6] proposed a unified predictive model by integrating clustering and classification techniques to provides a generalized solution for student employability prediction.

Apart from that, analysing massive online data has gained popularity among researchers recently through text mining techniques. Study done by [11] examined Business Intelligence (BI) and Big Data (BD) competencies by performing an automated content analysis of job ads through text mining approach. The findings reveal that the demand for BI competencies is still far bigger as they found three times more job ads than for BD competencies. According to [12], by using text mining approach, the study able to identify and characterize the skills and competencies currently required in the workplace which is needed to inventory their existing workforce skills in order to identify critical future human resources. Meanwhile, [13] applied text mining approach specifically Latent Dirichlet Allocation (DLA) topic model to identify skill sets from online job advertisements. [14] reported that Latent Semantic Indexing (LSI) model proposed in this study produces much finer and more timely readings of the labour market compare to the common official employment or job opening statistics which can be applied to different job markets, commercial sectors and geographical regions.

Employers these days are progressively worried about finding appropriate workers with specialized skills as well as furnished with high level state of employability skills and capacity to modify with fast changes in the business [15-16]. Online recruitment offers many benefits since many websites are being created, updated, and actively promoted and hence, produces massive amounts of vacancy data and provides volumes of potentially useful information that can be collected and analysed [17]. Digital

information has grown exponentially which resulted in sharp rise in demand for Data Scientist, Data Analyst, Data Artists and Data Visualiser vacancies, all over the world [18]. A report from the recent Digital Workforce of The Future by LinkedIn, which revealed an increase of 21% growth in demand in Malaysia comes from combination of skills encompassing Big Data, data analytics and web development [19]. Hence, this study will focus on a single segment which is analyst vacancies. It was also acknowledged that Analyst jobs are the most diversified, highly demanded and dynamic group, therefore it was particularly interesting for further exploration.

3. Methodology

This study adopted text mining approach by examining the content of job advertisements on analyst vacancy based on two types of information: job titles and job descriptions. Data published from Jobstore.com domain between May and June 2019 was retrieved using Import.io online app, a sample of 465 Analyst vacancies. An open source machine learning and data visualization, Orange software is used for data analysis purposes. The methodology comprises four phases namely; Text retrieval, text pre-processing, word cloud corpus generation and text clustering.

3.1. Text retrieval

The first stage in text retrieval involved obtaining the textual data. The data used for this study was retrieved from Jobstore Malaysia website, (https://www.jobstore.com/my/search-jobs). A sample post is shown in figure 1.

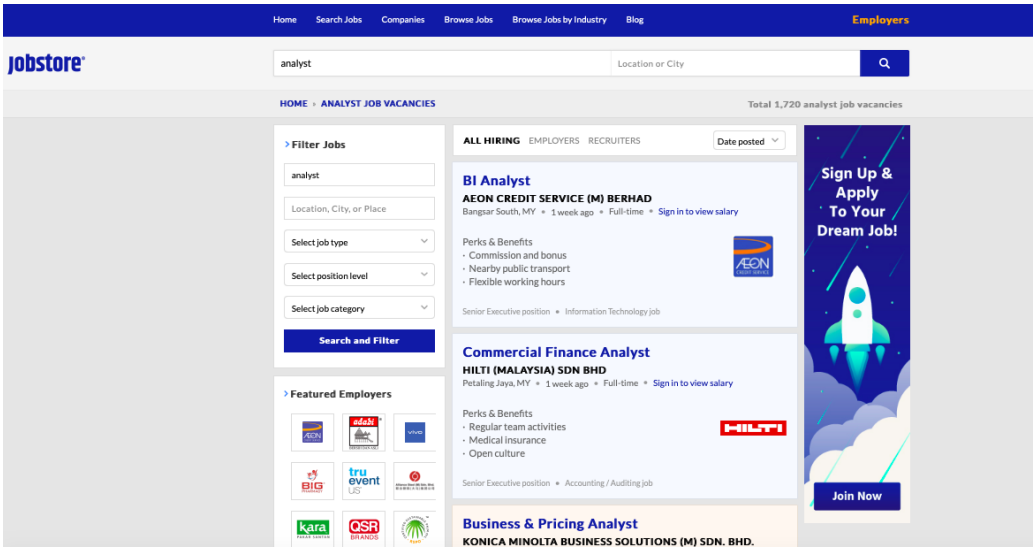


Figure 1. Sample of the document retrieve from the website.

This website contains vacancy and skill for analyst offered in Malaysia. This study used selected document regarding job title and description for analyst job posted from April 2019 until June 2019. The two months' time framework is chosen to capture the latest posting on job vacancies for analyst. A total of 465 documents were extracted from 1,720 analyst job vacancies.

3.2. Text Pre-processing

In Orange software, a widget called preprocess text is used to clean the dataset of analyst which obtained from online website (see figure 2). Preprocess text splits the text into smaller units known as tokens. Steps involved in the analysis are as follows:

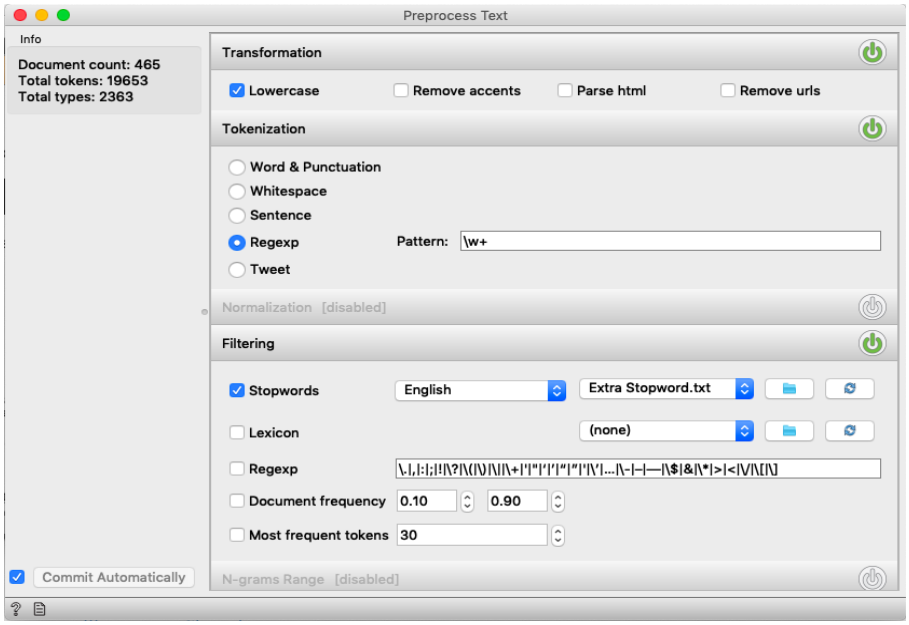


Figure 2. Preprocess Text parameters.

- ***Information on processed data***: 465 number of documents on the input, 19653 total number of tokens in the corpus and 2363 number of token types in the corpus.
- ***Transformation***: turn all text in the corpus to lowercase.
- ***Tokenization***: Regexp with \w+: matches only words is selected.
- ***Filtering***: Stopwords removes stopwords from text (e.g. removes 'and', 'or', 'in'...). In this study, Extra Stopword.txt file, own list of stopwords with one stopword per line is loaded to help remove unnecessary word.

3.3. Word Cloud Corpus Generation

Word cloud corpus can be generated with an add-ons option in Orange software. Word Cloud widget is connected after Preprocess Text widget in order to find the most frequent words in the text. Word Cloud displays tokens in the corpus together with the frequency of the word in corpus denoted as word size. A list of words presented by their frequency (weight) in the widget. The widget outputs documents, containing selected tokens from the word cloud which will be assigned with random colour.

3.4. Text Clustering

To perform text clustering, there are few steps needed to be carried out after cleaning the data using ***Preprocess Text*** widget (refer Figure 3 for further details). The ***Bag of Words*** model creates a corpus with word counts either absolute, binary (contains or does not contain) or sublinear (logarithm of the term frequency) for each data instance (document). In this study, parameter selected for bag of words model is Term Frequency: counting the number of occurrences of a word in a document. The ***Distances*** widget calculates distances between rows or columns in a dataset which is normalized to ensure equal treatment of individual features. The resulting distance matrix can be fed further to ***Hierarchical Clustering*** for uncovering groups in the data. The ***Distributions*** widget displays the value distribution of discrete or continuous attributes. If the data contains a class variable, distributions may be conditioned on the class. In this study, the graph displayed by the widget shows how many times (e.g., in how many instances) each attribute value appears in the data.

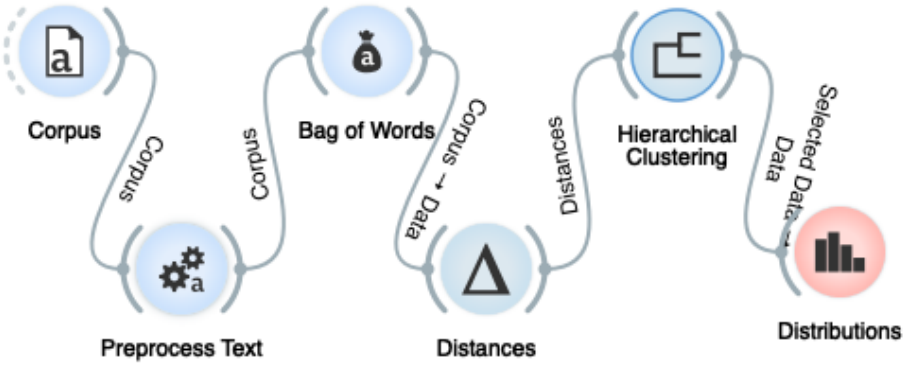


Figure 3. Text clustering workflow.

4. Vacancy Analysis with Orange

Following the process describe in section 3, this section presents the results of reliable text mining analysis on vacancy and skills for analyst in Malaysia.

4.1. Dataset Exploration

The results begin with visualization of the most frequent word from 465 document on job title and job description of unstructured data. The top eighteen words with its frequency value are shown in figure 4 and reference job positions (manager, senior, etc), analyst occupational area (business, accounting, financial, operation, support) and technologies (software, data, etc). Analysis from this attribute provides valuable insight into demand for analyst position and skills in Malaysia Labour market in 2019. Figure 5 adds more information from the content of job description analysis. It illustrates the importance of location, technology, responsibility and skill for a description of an analyst vacancy.

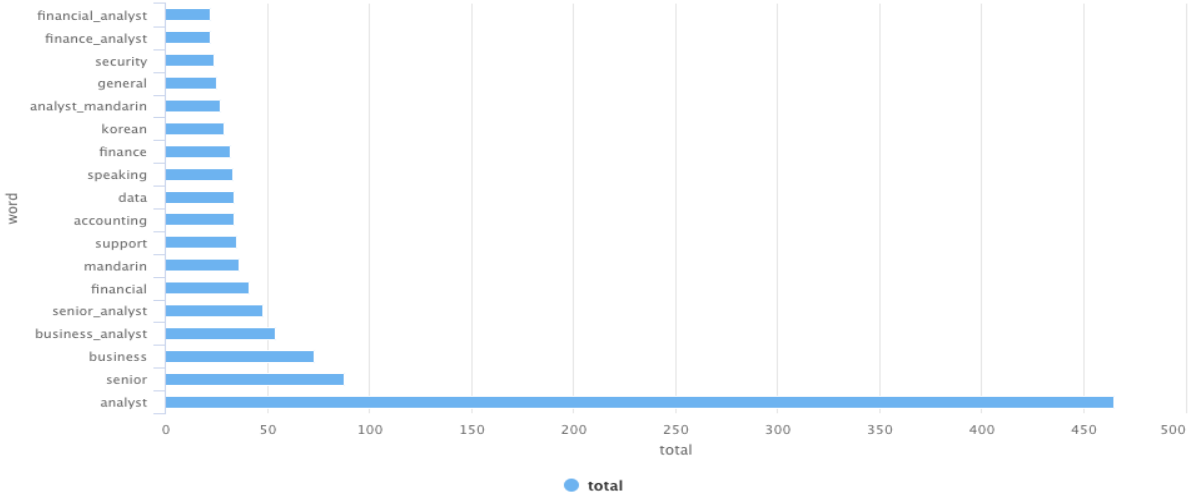


Figure 4. Top terms - Analyst dataset (Job Title).

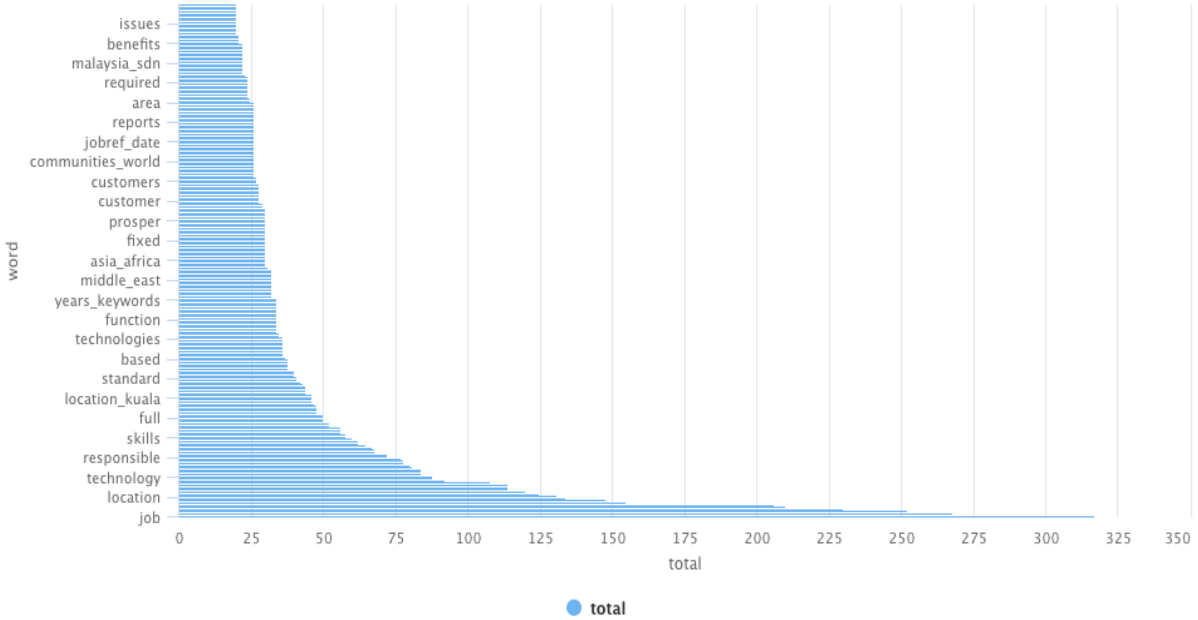


Figure 5. Top terms - Analyst dataset (Job Description).

4.2. Text Clustering Analysis

Orange for data mining supplies processes for computing distance between records in a dataset. Through comparison of textual attribute and similarity ranking construction, this study further analyze the document by identifying potential clusters of job vacancies. It was done by identifying rows potentially belonging to the same category . With similarity ranking from the distance measure, we can identify the closest and farthest records of the data sets that will form several homogenous within and heterogenoues between the cluster. Figure 6 and figure 7 illustrate outcome of hierarchical clustering that visualized the analyst job title into 5 major categories namely C1:Senior analyst, C2:Financial analyst, C3: Business analyst, C4: Accounting analyst and C5:Support analyst.

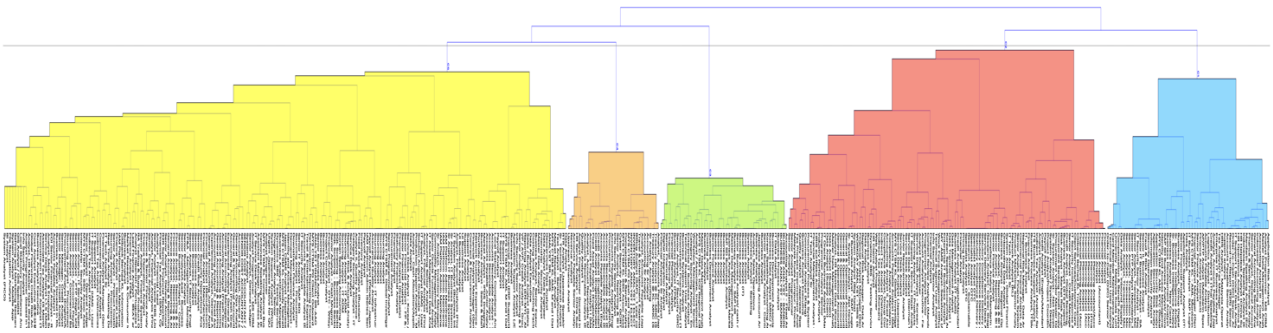


Figure 6. Hierarchical Clustering Output.

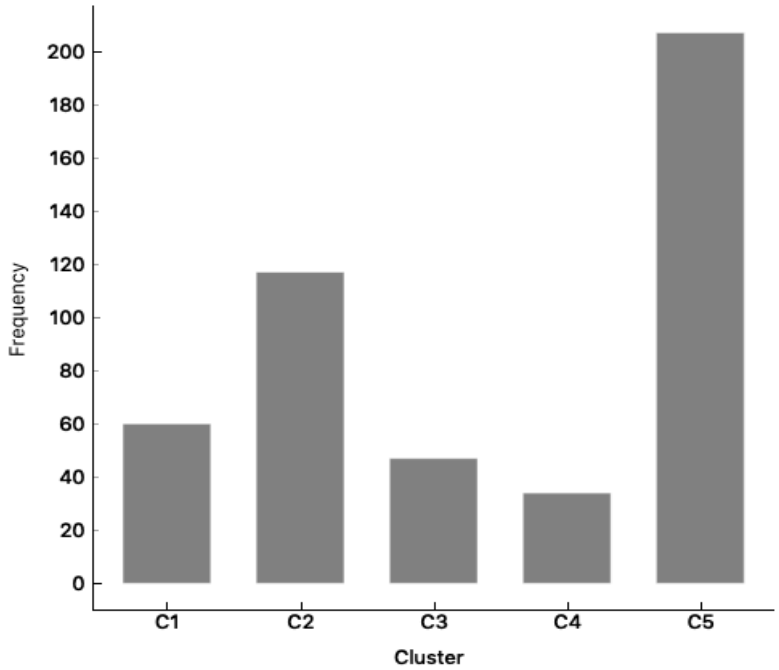


Figure 7. Distribution - Cluster Output.

Table 1 shows the details cluster of specific job title for each job analyst category. In this instance, we can visualize that the most demanded of job categories for analyst is the Support Analyst which also covers Senior analyst, Business analyst and Financial analyst. The least demanded analyst job category is the accounting analyst which only relates to accounting job information.

Table 1. Detailed Cluster of Analyst Job Title.

Cluster 1: Senior Analyst	Financial Planning Analyst (Accountant)-Priority	Cluster 4: Accounting Analyst
Accounts Payable Senior Analyst (China Support)	Financial Planning Analyst (Financial Reporting Accountant)	Accounting Sr Analyst
Finance & Senior Finance Analyst	Financial Planning Analyst/Accountant	Accounting Sr Analyst
General Accounting Senior Analyst (Indonesia Support)	Financial Planning Analyst (Accountant)-Priority	Accounts Payable Analyst (English/Mandarin/Korean/Vietnamese/Thai)
IT Security Senior Analyst - Regional	Payroll Analyst	Accounts Payable Analyst (English/Mandarin/Korean/Vietnamese/Thai)
Order-to-Cash (O2C) Senior Analyst (China Support)	Pricing Data Analyst	Accounts Payable Analyst I (Mandarin Speaking)
Order-to-Cash (O2C) Senior Analyst (China Support)	Procurement Process & System Analyst	Accounts Payable Senior Analyst (China Support)
Record-to-Report (R2R) Senior Analyst (China Support)	Procurement Vendor Master Data Analyst	Accounts Receivable Analyst (English/Mandarin/Korean/Vietnamese/Thai/Tagalog/Bahasa Indonesia)
Senior Analyst, General Accounting (GL & Fixed Assets)	Regional Risk Manager, Data Analyst	Accounts Receivable Analyst (English/Mandarin/Korean/Vietnamese/Thai/Tagalog/Bahasa Indonesia)
Senior Analyst, Transaction Monitoring	Regional Risk Manager, Data Analyst	Accounts Receivable Analyst (English/Mandarin/Korean/Vietnamese/Thai/Tagalog/Bahasa Indonesia)
Senior Analyst, Transaction Monitoring - NCB, FCSU	Risk Analyst - Market Risk Analytics (Traded)	Analyst Programmer (located in Warehouse, Shah Alam)
Senior Analyst, Transaction Monitoring - NCB, FCSU	Statutory Compliance Delivery Analyst	Analyst, Account Receivable (Mandarin Speaking)
Senior Analyst, Transaction Monitoring - NCB, FCSU	Statutory Compliance Delivery Analyst	Analyst, Accounts Payable (Japanese Speaking)
Senior Finance Analyst	Statutory Compliance Delivery Analyst	Analyst, Accounts Payable (Japanese Speaking)
Senior Financial Analyst	Tax Process Improvement Analyst	Billing Analyst (SAP) - 1 year contract (SSC)
Senior II, Business Analyst	Cluster 3: Business Analyst	Compensation & Benefit Analyst (Portuguese/Spanish)
Senior IT Business Analyst - SAP HRMS	Business Analyst - Banking	Credit Analyst (English/Mandarin/Korean/Thai/Vietnamese/Tagalog/Bahasa Indonesia)

Senior PTP Analyst (Mandarin_SAP)	Business Analyst - Banking	Credit Analyst (English/Mandarin/Korean/Thai/Vietnamese/Tagalog/ Bahasa Indonesia)
Sr. Business Intelligence Analyst	Business Analyst	Accounting Sr Analyst
Sr. Business Intelligence Analyst	Business Analyst	Accounting Sr Analyst
Accounts Payable Senior Analyst (China Support)	Business Analyst	Accounts Payable Analyst (English/Mandarin/Korean/Vietnamese/Thai)
Finance & Senior Finance Analyst	Business Analyst	Accounts Payable Analyst (English/Mandarin/Korean/Vietnamese/Thai)
General Accounting Senior Analyst (Indonesia Support)	Business Analyst	Accounts Payable Analyst I (Mandarin Speaking)
IT Security Senior Analyst - Regional	Business Analyst	Accounts Payable Senior Analyst (China Support)
Order-to-Cash (O2C) Senior Analyst (China Support)	Business Analyst	Accounts Receivable Analyst (English/Mandarin/Korean/Vietnamese/Thai/Tagalog/ Bahasa Indonesia)
Order-to-Cash (O2C) Senior Analyst (China Support)	Business Analyst	Cluster 5: Support Analyst
Record-to-Report (R2R) Senior Analyst (China Support)	Business Analyst	IT Asset Management Data Analyst
Senior Analyst, General Accounting (GL & Fixed Assets)	Business Analyst (Fp&A)	IT Security Analyst
Senior Analyst, Transaction Monitoring	Business Analyst (Global Markets)	IT Service Desk Analyst
Senior Analyst, Transaction Monitoring - NCB, FCSU	Business Analyst (It)	IT Support Analyst
Senior Analyst, Transaction Monitoring - NCB, FCSU	Business Analyst - Banking	IT Support Analyst
Senior Analyst, Transaction Monitoring - NCB, FCSU	Business Analyst - Sap Hr	IT Support Analyst
Senior Finance Analyst	Business Data Analyst (Mnc/ Telco) @Kl	IT Support Analyst (IAM) - User Admin
Cluster 2: Financial Analyst	Business Development Analyst | Internship	IT Support Analyst (IAM) - User Admin
Finance Analyst	Business Development Associate / Analyst	IT System Analyst (Business Intelligence)
Financial Analyst	Business Process Analyst	IT Tech Analyst Sr
Financial Analyst	Business Systems Analyst	SQL Scripting - Business Banking Analyst (Renewable Contract)
Financial Analyst	Category Sourcing Analyst - Engineering/ Refinery/ Marketing	SQL Scripting - Business Banking Analyst (Renewable Contract)
Financial Analyst	Costing Analyst	Senior Oracle Functional Analyst
Financial Analyst	Digital Integration Analyst	Senior Programmer Analyst
Financial Analyst (Accountant)	Enterprise It Business Analyst	Senior Security Operations Analyst
Financial Analyst Manager (General Insurance)	MYS CTSM KL ICG Regional CBCM And Governance Analyst - Vice President	Service Desk Analyst
Financial Management Analyst (Data Analytics)	Master Data Management Analyst (English/Mandarin/Korean)	Support Analyst
Financial Planning Analyst	Senior Analyst, It Security Grc	System Analyst (FICO)
Financial Planning Analyst (Accountant)	Senior Business Analyst - Controlling	System Analyst - Mobile App
Financial Planning Analyst (Accountant)	Senior Business Analyst/ Cash Management SME-- Only Local Malaysians	System Support Analyst (Wintel)

4.3. Word Cloud

A word cloud is a graphical representation of word frequency. The word cloud created using Orange software for job description in reference to analyst job categories are shown in the following figure. All documents were passed into an online word-cloud generator and the words are displayed in figure 8 to figure 12. Analysis of unstructured data on job description for business Analyst category reveals that “support”, “sourcing”, “services” and “data” are being emphasized in discussions about Business Analyst vacancy. Words that are frequently used to describe the Accounting analyst job descriptions are “clients”, “account” and “reporting” (figure 9).



Figure 8. Top similarity term of job description for Business Analyst.



Figure 9. Top similarity term of job description for Accounting Analyst.

Meanwhile, figure 10 displayed the words that are frequently used to describe the Support Analyst job descriptions are “sourcing”, “company”, “service” and “data”. Figure 11 displayed the words that are frequently used to describe the Financial Analyst job descriptions are “reporting”, “client” and “company”. While in the Senior Analyst category (see figure 11), the frequent words are “business”, “financial” and “support”.



Figure 10. Top similarity term of job description for Support Analyst.


accounting
finance
client
location
reporting
financial
business
description
company
senior
position
planning
enterprise
responsible
services
hewlett
salary
data
subsidiary
analysis
technology
support
general
role
management
provide
accountant
singapore
opportunity
finland
statement
insurance
hong
lumpur
malaysia
manager
quality
kuala
packard
local
people
global
group
month
preparation
society
organization
high
name
bring
compliance
permanent
key
positions
performance
processes
together
intercompany
budgeting
possibilities
meaningful
manufacturing
technologies
analysts
billing
introduction
knowledge
reputable
ensuring
good
years
work
looking
leading
solutions
ifrs
part
brings
degree
collating
challenge
provider
industry
process
timely
ibm
system
entity
hiring
delivery
execute
plan
listed
assets
south
new
within
digital
reviewing
preparing
fact
type
level
lead
china
leader
accurate
control
kong
responsible
mines
businesses
driving
materials
develop
end
uae
opening
building
pricing
reconciliation
accenture
keywords
function
experience
operations
summary
selangor
cost
clients
creates
door
needs
play
costing
functions
regional
banking
enables
solve
fixed
various
deliverables
bank
career
subang
skills
closing
impact
ensure
join
team
full
operational
project
non
jaya
next
month
group
preparation
society
people
global
organization
high
name
packard
local
role
company
solutions
leading
hiring
delivery
work
looking
provide
market
years
good
knowledge
reputable
ensuring
known
manager
malaysia
excellent
east
forward
compliance
quality
kuala
bring
junior
users
accounts
close
oracle
innovation
incumbent

5. Conclusion

This study analyses job market demands particularly analyst job vacancies which provides directly actionable information about vacancy and skills demand. The findings show the classification results and enhance information extraction from job descriptions. By accessing publicly available vacancy data, with web and text mining tools, the study able to extract valuable facts about competences and abilities sought by employers. This approach supports skills research by providing a fact-based alternative for the resource-expensive employer surveying. However, the findings extract mostly on vacancy information and lacking of skills information. The findings provide occupational analysis which is lacking a reflection of current needs, and leads to the gap between skills demand and supply. This limitation is due to the data obtained from online job posts which mostly display the jobs-related information. Although this study only focus on specific job direction, the steps taken could be applied to any type of job vacancies. Future directions in this work include more datasets from several online job websites and also more advanced text pre-processing and cleaning techniques to further improve the classification results. Furthermore, future works also will involves comparing and combining university skills and industry work skills in order to minimize the imbalance of industry-university skills demand. Therefore, the work will be extended by considering predictive model in order to map the university-industry skills and demands from both parties.

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